this would nurture a sense that all the buttons and toggles work together in a coordinated and unified manner. A similar feeling prevails among string theorists. Results found over many years of research are dovetailing and converging. This has instilled a growing confidence among researchers that string theory is closing in on one powerful, coherent framework— which has yet to be unearthed fully, but ultimately will expose nature's inner workings with unsurpassed clarity and comprehensiveness.

In recent times, nothing illustrates this better than the realization that sparked the *second superstring revolution*—a revolution that has, among other things, exposed another hidden dimension entwined in the spatial fabric, opened new possibilities for experimental tests of string theory, suggested that our universe may be brushing up against others, revealed that black holes may be created in the next generation of high-energy accelerators, and led to a novel cosmological theory in which time and its arrow, like the graceful arc of Saturn's rings, may cycle around and around.

The Second Superstring Revolution

There's an awkward detail regarding string theory that I've yet to divulge, but that readers of my previous book, *The Elegant Universe*, may recall. Over the last three decades, not one but *five* distinct versions of string theory have been developed. While their names are not of the essence, they are called Type I, Type IIA, Type IIB, Heterotic-O, and Heterotic-E. All share the essential features introduced in the last chapter —the basic ingredients are strands of vibrating energy—and, as calculations in the 1970s and 1980s revealed, each theory requires six extra space dimensions; but when they are analyzed in detail, significant differences appear. For example, the Type I theory includes the vibrating string loops discussed in the last chapter, so-called *closed strings*, but unlike the other string theories, it also contains *open strings*, vibrating string snippets that have two loose ends. Furthermore, calculations show that the list of string vibrational patterns and the way each pattern interacts and influences others differ from one formulation to another.